The Rational Root Theorem
Answer Key

Algebra 2

Quick Answers

1. (b) the value at x = -1 = 0, —
2. 0
3. -1, 2, 3
4. -1, -3, 2
5. (b) the value at x = -2 = 0, —
6. $(x+2)(3x-2)(x-1)$
7. 3, $\frac{1}{2}$, -1
8. the value at x = 2 = 0, the value at x = 4 = 6, —
9. -1, 2, 3, -2
10. $(x+1)(3x-2)(2x+1)$
11. 3, $-1+\sqrt{3}$, $-1-\sqrt{3}$
12. (a) the rational zero = 1, (b) the remaining zeros = $2+i$, $2-i$, —

What is verified

14 of 18 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 1, 5, 8, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2
HSA-APR.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 18 tagged checks.

1. $f(x) = x^3 - 4x^2 + x + 6$. (a) List the possible rational zeros. (b) Find $f(-1)$.

Part (a) — instructor-judged. Numerators divide the constant term 6: $1, 2, 3, 6$. Denominators divide the leading coefficient 1, so only 1. The complete list is $\pm1, \pm2, \pm3, \pm6$ — eight values. The usual omission is the negatives; deduct for a list that drops them, and for any value that does not divide 6.

Solution (b): $(-1)^3 - 4(-1)^2 + (-1) + 6 = -1 - 4 - 1 + 6$. Since the value is 0, -1 is a zero and $x+1$ is a factor. $\boxed{f(-1) = 0}$

2. $g(x) = 2x^3 - 3x^2 - 8x - 3$. Find $g(3)$.

Solution: $2(27) - 3(9) - 8(3) - 3 = 54 - 27 - 24 - 3$. The value is 0, so 3 is a zero of g. $\boxed{g(3) = 0}$

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3. Find all zeros of $f(x) = x^3 - 4x^2 + x + 6$.

Solution: Problem 1 established that -1 is a zero. Dividing by $x+1$ (synthetic division with -1 on $1, -4, 1, 6$) gives the quotient $x^2 - 5x + 6$, which factors as $(x-2)(x-3)$. A cubic has three zeros and all three are here. $\boxed{x = -1,\ 2,\ 3}$

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4. Find all zeros of $f(x) = x^3 + 2x^2 - 5x - 6$.

Solution: Candidates are $\pm1, \pm2, \pm3, \pm6$. Testing $x = -1$: $-1+2+5-6 = 0$. Dividing by $x+1$ leaves $x^2 + x - 6$, which factors as $(x+3)(x-2)$. $\boxed{x = -1,\ -3,\ 2}$

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5. $h(x) = 3x^3 + x^2 - 8x + 4$. (a) List the possible rational zeros. (b) Find $h(-2)$.

Part (a) — instructor-judged. Numerators divide 4: $1, 2, 4$. Denominators divide 3: $1, 3$. The complete list is $\pm1, \pm2, \pm4, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}$. Full credit needs the fractional candidates as well; dropping them is the error this problem exists to expose, and it is why problem 6 has a zero of $\frac{2}{3}$.

Solution (b): $3(-8) + 4 + 16 + 4 = -24 + 24$. The value is 0, so -2 is a zero.

$\boxed{h(-2) = 0}$

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6. Factor completely: $3x^3 + x^2 - 8x + 4$.

Solution: From problem 5, -2 is a zero, so $x+2$ is a factor. Synthetic division with -2 on $3, 1, -8, 4$ gives $3, -5, 2$ and remainder 0, so the quotient is $3x^2 - 5x + 2$. That factors as $(3x-2)(x-1)$. $\boxed{(x+2)(3x-2)(x-1)}$

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7. Find all zeros of $f(x) = 2x^3 - 5x^2 - 4x + 3$.

Solution: Candidates are $\pm1, \pm3, \pm\frac{1}{2}, \pm\frac{3}{2}$. Testing $x = 3$: $54-45-12+3 = 0$. Dividing by $x-3$ leaves $2x^2 + x - 1$, which factors as $(2x-1)(x+1)$, giving a fractional zero and a whole one. $\boxed{x = 3,\ \frac{1}{2},\ -1}$

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8. $q(x) = x^3 - 6x^2 + 11x - 6$. Find $q(2)$ and $q(4)$, then say which is a zero.

Solution: $q(2) = 8 - 24 + 22 - 6$. $\boxed{q(2) = 0}$

And $q(4) = 64 - 96 + 44 - 6$. $\boxed{q(4) = 6}$

Written part — instructor-judged. The answer must name 2 as the zero and give the reason: substituting 2 made the polynomial equal 0, while 4 gave a non-zero value, and a number is a zero exactly when the polynomial evaluates to 0 there. Naming 2 with no reference to the computed values does not earn the mark.

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9. Find all zeros of $p(x) = x^4 - 2x^3 - 7x^2 + 8x + 12$.

Solution: Candidates are $\pm1, \pm2, \pm3, \pm4, \pm6, \pm12$. Testing $x = -1$ gives $1 + 2 - 7 - 8 + 12 = 0$, so divide by $x+1$; the quotient is $x^3 - 3x^2 - 4x + 12$, which groups as $x^2(x -$

$3) - 4(x-3) = (x-3)(x-2)(x+2)$. A quartic has four zeros. $\boxed{x = -1,\ 2,\ 3,\ -2}$

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10. Factor completely: $6x^3 + 5x^2 - 3x - 2$.

Solution: Numerators divide 2 and denominators divide 6, so the candidate list is long — start with the whole numbers. Testing $x = -1$: $-6 + 5 + 3 - 2 = 0$, so $x + 1$ is a factor. Synthetic division with -1 on $6, 5, -3, -2$ gives $6, -1, -2$, so the quotient is $6x^2 - x - 2$, which factors as $(3x-2)(2x+1)$. $\boxed{(x+1)(3x-2)(2x+1)}$

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11. Find all zeros of $f(x) = x^3 - x^2 - 8x + 6$.

Solution: Candidates are $\pm1, \pm2, \pm3, \pm6$. Only $x = 3$ works: $27 - 9 - 24 + 6 = 0$. Dividing by $x - 3$ leaves $x^2 + 2x - 2$, which has no rational roots, so finish with the quadratic formula: $x = \dfrac{-2 \pm \sqrt{4+8}}{2} = -1 \pm \sqrt{3}$. The other two zeros are irrational, which is exactly why no candidate on the list found them. $\boxed{x = 3,\ -1+\sqrt{3},\ -1-\sqrt{3}}$

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12. $f(x) = x^3 - 5x^2 + 9x - 5$. (a) The rational zero. (b) The remaining zeros. (c) Why the theorem could not find them.

Solution (a): Candidates are $\pm1, \pm5$. Testing $x = 1$: $1 - 5 + 9 - 5 = 0$. $\boxed{x = 1}$

Solution (b): Dividing by $x - 1$ leaves $x^2 - 4x + 5$, whose discriminant is $16 - 20 = -4$, so the remaining zeros are non-real: $x = \dfrac{4 \pm 2i}{2}$. $\boxed{x = 2 \pm i}$

Part (c) — instructor-judged. A full-credit explanation says the theorem only ever proposes *rational* numbers, so a zero that is not rational — and these two are not even real — can never appear on its candidate list. It should add that the theorem is a first step that lowers the degree, after which the leftover quadratic is handled by the quadratic formula. Restating the two zeros with no comment on why the theorem missed them earns nothing.